Worksheet 5B: Right Triangle Trigonometry — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: SOH-CAH-TOA

1. Opp=3, Adj=4, Hyp=5; $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$
2. $\sin\theta = \frac{5}{13}$, $\cos\theta = \frac{12}{13}$, $\tan\theta = \frac{5}{12}$
3. $\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$
4. $\sin\theta = \frac{24}{25}$, $\cos\theta = \frac{7}{25}$, $\tan\theta = \frac{24}{7}$
5. (a) $\sqrt{2}$ (b) $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{2}}{2}$, 1
6. (a) $1, \sqrt{3}, 2$ (b) $\frac{1}{2}$, $\frac{\sqrt{3}}{2}$, $\frac{1}{\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2}$, $\frac{1}{2}$, $\sqrt{3}$
7. (a) 0.4226 (b) 0.3090 (c) 1.3270
8. (a) 36.87° (b) 36.87° (c) 68.20°

Part B: Finding Missing Sides

9. 5.74 cm
10. 10.46 m
11. 8.01 m
12. 17.66 cm
13. 17.14 m
14. 8.98 m
15. 13.53 m
16. (b) $\approx$ 1,028 km
17. 1.93 mm
18. 28.68 mm
19. (a) 4.76° (b) 7.21 m

20. (a) 3.43° (b) 30 m
21. (a) 64° from horizontal (b) $\approx$ 1.08 m

Part C: Finding Missing Angles

22. 22.6°
23. 61.9°
24. 35.0°
25. 53.1°
26. 61.4°
27. (a) 1.30 parsecs (b) 4.24 light-years
28. (a) 100 parsecs (b) Angle too small to measure accurately
29. (a) 48.6° (b) 41.8°
30. (a) 38.1° (b) Light must hit at angles greater than critical angle
31. (a) 15.9° (b) 33.1°
32. 0.356 nm
33. Would require $\sin\theta > 1$, which is impossible

Part D: Angles of Elevation and Depression

34. 107.2 m
35. (a) 189.5 m (b) 30.0 m
36. 307.8 m
37. (a) 0.00708° (b) $\approx$ 54 million km
38. $\approx$ 33,500 km
39. (a) $\approx$ 6.5 mm radius
40. (a) $\approx$ 0.87 mm
41. (a) $\approx$ 62 nm (b) Wafer must be very flat
42. (a) $\approx$ 22 nm (b) Factor of 2.8 smaller
43. (b) $\approx$ 452 m (c) $\approx$ 421 m
44. (b) $\approx$ 46.1° or 0.804 rad

—